Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/02/2024
PI Name:	Jaiswal, Pankaj Ph.D.		
Project Title:	Alternative Splicing and Transcriptome Modulation Under Space Flight Response in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jaiswalp@oregonstate.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	541-737-8471
Organization Name:	Oregon State University		
PI Address 1:	Department of Botany and Plant Pathology		
PI Address 2:	2082 Cordley Hall, 1500 SW Jefferson Street		
PI Web Page:			
City:	Corvallis	**State:**	OR
Zip Code:	97331	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	04/01/2022	**End Date:**	03/31/2024
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	O'Rourke, Aubrie	**Contact Phone:**	
Contact Email:	aubrie.e.orourke@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2024 per NSSC information (Ed., 2/23/23)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	80NSSC22K0855		
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Task Description:	The space flight transcriptome studies collected by the NASA GeneLab and analyzed by its Plant Analysis Working Group (PlantAWG) have facilitated analysis and tool development around profiling the differentially expressed genes (DEG), enrichment of pathways, and Gene Ontology-based (GO-based) functional annotation. To explore additional novel molecular events from the same GeneLab studies, we will (1) analyze the alternatively spliced (AS) forms of transcripts (RNA molecules) identified from the ground control (GC) and the space flight (FLT) samples; (2) identify mutations (if any) gained due to cosmic radiation in the transcribed regions of the genome; and (3) diagnose the AS and somatic mutation consequences on the structure and function of the genes. Our data analysis and collaborative work will extend the current functionality of the NASA GeneLab data by adding alternative splicing, novel transcribed regions and genes, and any functional significance, thus providing researchers insights into the modulated transcriptome's structural and functional consequences in response to space flight. The project workflows will help improve the reference genome annotation of Arabidopsis and set up a platform for analyzing current and future data from non-plant eukaryotes like humans, mice, yeast, and other models. In order to develop a deeper understanding of the causal effects of alternative splicing (AS) in the model plant Arabidopsis thaliana and other space travel-bound plant species, we propose to build an AS Atlas for plants in response to microgravity. The AS atlas will analyze the core set of NASA GeneLab plant studies and put forth a plan to build and extend resources for current and future studies on plants and other eukaryotes, including humans and mouse models. Our data analysis and collaborative work will extend the differential gene expression tools and functional enrichment analysis tools currently provided by the GeneLab data. The identification of novel AS sites and genes is anticipated to provide candidate genes to target for developing smart crops and synthetic biology-based approaches to improve adaptation to microgravity. The project-specific aims include: 1. Develop an alternative splicing atlas of Arabidopsis genes 2. Identify novel genetic mutations gained in the transcribed regions of the genome during space flight (FLT) 3. Validate alternative splicing events, subject to the availability of biological material sampled from the existing GeneLab studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project explores the identification of novel alternative spliced isoforms of Arabidopsis thaliana genes, mutations acquired, and downstream effects on the structure and function of the genes transcribed in response to space flight and microgravity. By analyzing the data from the GeneLab data portal, the project will extend the functionality limited by the existing differential gene expression and Gene Ontology enrichment analysis. The analyzed data will provide researchers additional insights into the modulated transcriptome's structural and functional consequences in response to space flight, hardware and growth conditions. The identification of novel AS sites and genes is anticipated to provide candidate genes to target for developing smart crops and synthetic biology-based approaches to improve adaptation to microgravity. The proposed development of methodologies and data analytics workflows may also help analyze the current and future data from plant and non-plant eukaryotes like humans, mice, yeast, and other models.
Task Progress:	We analyzed the transcriptome data available from NASA GeneLab studies OSD-37, OSD-38, OSD-120, OSD-251, OSD-314. The analysis included (1) exploring the alternative splicing events in the ground control and flight samples of plants used in these experimental studies for the protein coding genes, and (2) identifying Single Nucleotide Polymorphism (SNP) mutations found in these samples that may be a result of the respective growth condition exposures. We found about 6600 protein coding genes that undergo splicing and show differential gene expression. These genes are enriched in functions such as DNA repair, response to radiation, response to light stimulus, cellular responses to stress. Flight samples in almost all the studies show more number of unique SNP mutations gained in the microgravity environment of the space. The data sets are getting ready for publication. We have published four peer-reviewed articles, given ten talks, presented three posters, and submitted one white paper. Our project team member was also selected for NASA GeneLab Plant Analysis Working Group (AWG) member spotlight.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Articles in Peer-reviewed Journals	Gupta P, Elser J, Hooks E, D'Eustachio P, Jaiswal P, Naithani S. "Plant Reactome Knowledgebase: empowering plant pathway exploration and OMICS data analysis." Nucleic Acids Research. 2024 Jan 5;52(D1):D1538-47. https://doi.org/10.1093/nar/gkad1052 ; PMID: 37986220, PMCID: PMC10767815 , Jan-2024
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